can be absorbed as titanium tritide on an area of 5 cm^2. The target is cooled by water to minimize tritium evaporation. The passage of the beam through the target also results in the emission of electrons which are removed by attraction to an anode to avoid interference with the beam.

With a voltage of 100 kV and an ion current of 0.5 mA, this accelerator can produce approximately 10^{10} n s^{-1} with an energy of 14 MeV. It is often desirable to reduce the kinetic energy of these neutrons to thermal values since thermal neutrons have a much larger probability of interacting with target nuclei. Moderation of the neutrons to thermal energies ($\approx$0.025 eV) can be accomplished by placing water or paraffin around the target (see §§7.6 and 19.5). The flux of the thermal neutrons is relatively low, on the order of 10^8 n cm^{-2} s^{-1}. However, the production rate of neutrons increases as the beam energy and beam current increases.

Small accelerators of similar design are commercially available with such dimensions that they can be lowered into bore-holes for in situ neutron activation analysis during gas or oil prospecting.

More advanced designs of this type of accelerator use coupling of transformers, rectifiers, and condensers to increase the voltage. With the larger single-stage accelerators it is possible to obtain beams of protons and deuterons which have energies of several MeV and currents of about 10 mA.

16.4. VAN DE GRAAFF ACCELERATORS

A type of electrostatic accelerator that has been of great value to nuclear science since its invention is that designed by van de Graaff in 1931. The van de Graaff accelerator (VdG) can provide beams of higher energy than the single-stage Cockcroft-Walton accelerators with a very precisely defined energy. In a modification known as the tandem-VdG beams of 20 MeV protons and 30 MeV α-particles can be achieved. Positive ions of higher Z and electrons can also be accelerated in VdG machines.

The principle of the VdG accelerator is illustrated in Figure 16.3. A rapidly moving belt of a non-conducting material such as rubberized fabric accumulates positive charge as it passes an array of sharp spray points which transfer electrons from the belt to the spray points. The positive charge on the belt is continuously transferred by the movement of the belt away from ground. At the high-voltage terminal, a hollow metal sphere or nearly cylindrical shape, another set of spray points neutralize the charges on the belt by electrons emitted from the spray points. The result is the transfer of the positive charge to the sphere. By a continuous process of transfer of positive charge to the sphere via movement of the belt a high potential relative to ground can be built up on the sphere. The limit of the voltage that can be accumulated on the hollow electrode is determined by its discharge potential to the surrounding housing. If it is insulated by some pressurized non-conducting gas such as N_2, CO_2, or SF_6, potentials of $\sim$15 MV can be

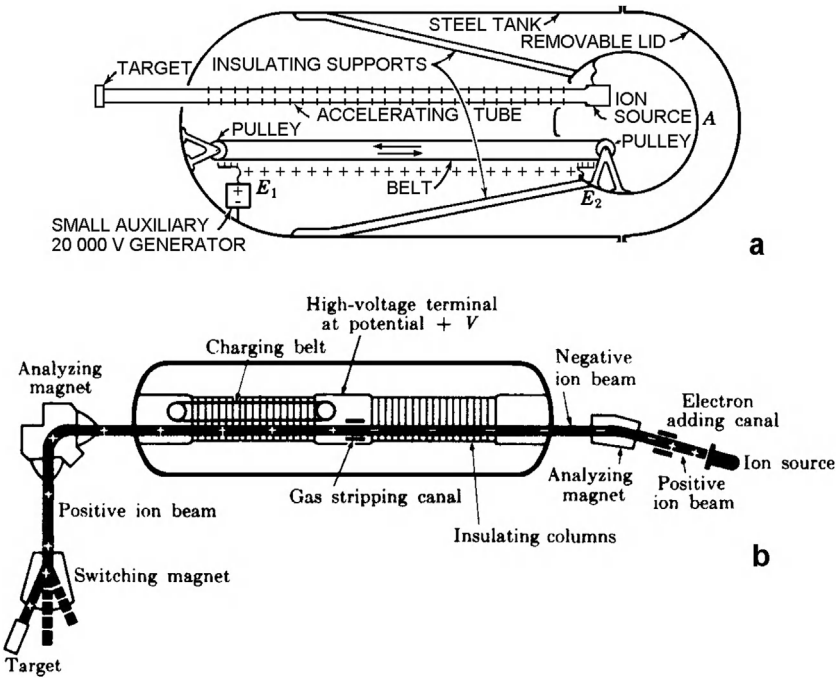


Figure 16.3 (a) The main principle of the van de Graaff accelerator. (According to R. S. Shankland.) (b) The principle of the tandem-VdG accelerator. *(According to R. V. van de Graaff.)*

achieved. This potential can be used to accelerate protons to energies of ~15 MeV in a single stage.

The single-stage VdG has only one accelerating tube. In the tandem VdG modification, atoms are first given a negative charge by bombardment with low energy electrons in the ion source, and then accelerated from ground to the positive potential at the center electrode. At this stage, in the center of the accelerator, the negative ions are stripped of electrons by passage through a thin foil or a gas volume to become positive ions. The average charge, z, of an ion of velocity v (m/s) and nuclear charge Z emerging from a solid material is given by the following empirical relation.

$$z = Z\left[1 + \left\{v/\left(3.6 \times 10^6 (\mathrm{m/s})\ Z^{0.45}\right)\right\}^{-1.67}\right]^{-0.6} \tag{16.3}$$

In the second stage of acceleration, the *post-stripper*, the positive ions are accelerated as they move from the positive potential on the center electrode back to ground. The emergent beam has a total energy of 20 MeV if the accelerating potential is 10 MV. Beams of hydrogen, helium, nitrogen, oxygen, and heavier ions can be accelerated similarly. The highest energy so for achieved in a tandem VdG accelerator is 60 MeV corresponding to M^- ions in the first stage and M^{5+} in the second ($1 \times 10 + 5 \times 10 =$

60 MeV). By using two separate tanks and alternatingly positively and negatively charged ions, three-stage VdG machines with maximum proton energies up to ~45 MeV have been built.

The energy of a beam produced by a VdG generator is extremely precise. However, the current intensity, 10–100 μA, is somewhat less than that of other accelerators. The beam current i (amperes) is given by the relation

$$i = q\, I_o = \mathbf{e}\, z\, I_0 \tag{16.4}$$

where q is the projectile charge (C), I_0 the incident particle current (particles s^{-1}), $\mathbf{e} = 1.6 \times 10^{-19}$ C, and z the net charge (in electron units) of the beam particle (ion).

16.5. MULTIPLE-STAGE LINEAR ACCELERATORS

The potential obtained from a high voltage generator can be used repeatedly in a multiple-stage accelerator process ($n > 1$ in (16.1)). The linear accelerator operates on this principle. The accelerator tube consists of a series of cylindrical electrodes called drift tubes. In the Wideröe linear accelerator, the electrodes are coupled to a radio frequency generator in the manner shown in Figure 16.4. The high voltage generator gives a maximum voltage V which is applied to the electrodes by the radio frequency so that the electrodes alternate in the sign of the voltage at a constant high frequency. If the particles arrive at the gap between electrodes in proper phase with the radio frequency such that the exit electrode has the same charge sign as the particle and the entrance electrode the opposite, the particles are accelerated across the gap. Each time the particles are accelerated at an electrode gap they receive an increase in energy of qV; for n electrode gaps the total energy acquired is nqV (16.1). Inside the drift tubes no acceleration takes place since the particles are in a region of equal potential.

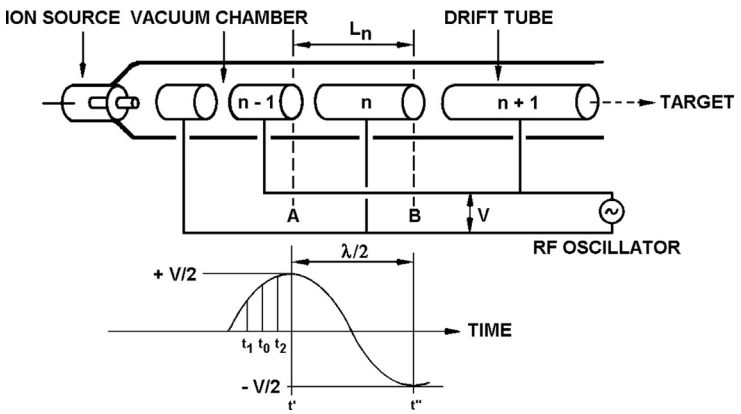


Figure 16.4 Principle of a multi-stage high voltage linear accelerator of Wideröe type.